Geometry Angles & Parallel Lines Cheat Sheet

Unit Focus: *angle relationships and transversals*

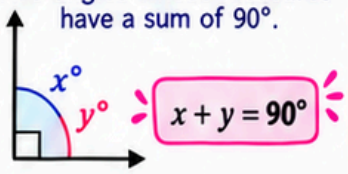
1 KEY VOCABULARY

transversal	A line that intersects two or more lines (often parallel lines) at different points.
corresponding angles	Angles in the same relative position at each intersection of a transversal and two lines.
alternate interior angles	Angles on opposite sides of the transversal and between the two lines.
alternate exterior angles	Angles on opposite sides of the transversal and outside the two lines.
same-side interior angles	Angles on the same side of the transversal and between the two lines.
vertical angles	Opposite angles formed by two intersecting lines. They are congruent.
linear pair	Two adjacent angles whose noncommon sides form a straight line. They are supplementary (sum to 180°).

2 BASIC ANGLE RELATIONSHIPS

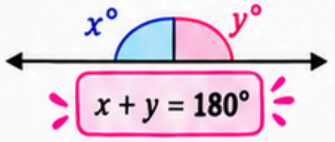
Complementary Angles

Two angles whose measures have a sum of 90°.



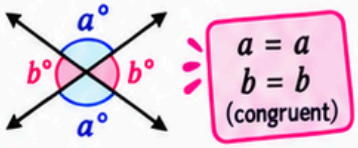
Supplementary Angles

Two angles whose measures have a sum of 180°.



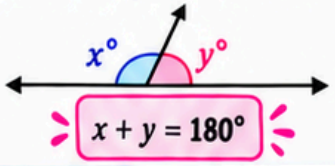
Vertical Angles

Opposite angles formed by two intersecting lines are congruent.



Linear Pair

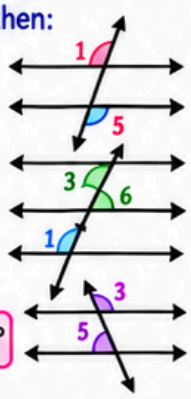
Two adjacent angles whose noncommon sides form a straight line are supplementary.



3 PARALLEL LINES CUT BY A TRANSVERSAL

If two parallel lines are cut by a transversal, then:

Corresponding angles are congruent (the same measure).	$\angle 1 \cong \angle 5$
Alternate interior angles are congruent (the same measure).	$\angle 3 \cong \angle 6$
Alternate exterior angles are congruent (the same measure).	$\angle 1 \cong \angle 8$
Same-side interior angles are supplementary (add to 180°).	$\angle 3 + \angle 5 = 180°$



4 SOLVING ANGLE EQUATIONS

Example: Corresponding angles $(3x + 15)°$ and $(5x - 25)°$ are congruent. Find the value of x and each angle.

Step 1: Set the angles equal.

$$3x + 15 = 5x - 25$$

Step 2: Solve for x.

$$40 = 2x$$
$$x = 20$$

Step 3: Find each angle.

$$3(20) + 15 = 75°$$
$$5(20) - 25 = 75°$$

Answer: $x = 20$. Each angle is 75°.

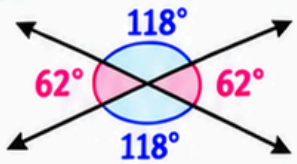
5 IDENTIFY THE ANGLE PAIR

Use the positions of the angles to identify the relationship.

Corresponding angles	In the same relative position at each intersection (one at the top, bottom, left, or right in the matching corners).
Alternate interior angles	Inside the two lines and on opposite sides of the transversal.
Alternate exterior angles	Outside the two lines and on opposite sides of the transversal.
Same-side interior angles	Inside the two lines and on the same side of the transversal (they are supplementary).

6 VERTICAL ANGLES EXAMPLE

If one angle is 118°, its vertical angle is also 118°, and each adjacent angle is 62°.



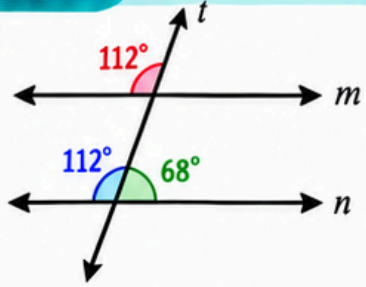
- Vertical angles are congruent.
- Each adjacent angle forms a linear pair with 118°, so its measure is 180° − 118° = 62°.

Answer: Vertical angle = 118°. Each adjacent angle = 62°.

7 USE WITH A DIAGRAM

In the diagram, lines m and n are parallel and cut by transversal t. If one angle is 112°, find the indicated angles.

- Corresponding angle: 112°
- Alternate interior angle: 112°
- Same-side interior angle: 68° (because 112° + 68° = 180°)



8 QUICK REMINDERS & COMMON MISTAKES

DO REMEMBER:

- Corresponding angles are congruent only when the lines are parallel.
- Alternate interior angles are congruent (when parallel).
- Alternate exterior angles are congruent (when parallel).
- Same-side interior angles add to 180° (when parallel).
- Vertical angles are always congruent.
- A linear pair is supplementary (180°).

COMMON MISTAKES:

- Corresponding angles are not congruent if the lines are not parallel.
- Same-side interior angles are not congruent (they are supplementary).
- Do not confuse alternate interior angles with alternate exterior angles.
- Angle relationships only apply when the two lines are parallel (unless using vertical angles or a linear pair).
- Be careful with angle labeling and make sure you're comparing the correct angles.

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Geometry Angles & Parallel Lines Cheat Sheet

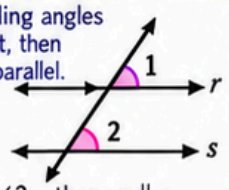
More Practice: proving lines parallel and angle theorems

1 CONVERSES TO PROVE LINES PARALLEL

If certain angle relationships are true, then the lines are parallel (converse statements).

Corresponding Angles Converse

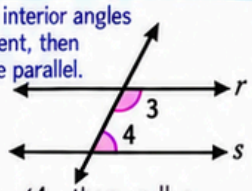
If corresponding angles are congruent, then the lines are parallel.



If $\angle 1 \cong \angle 2$, then $r \parallel s$.

Alternate Interior Angles Converse

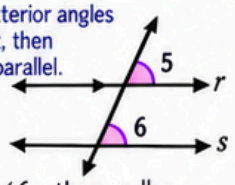
If alternate interior angles are congruent, then the lines are parallel.



If $\angle 3 \cong \angle 4$, then $r \parallel s$.

Alternate Exterior Angles Converse

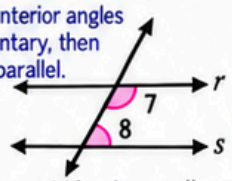
If alternate exterior angles are congruent, then the lines are parallel.



If $\angle 5 \cong \angle 6$, then $r \parallel s$.

Same-Side Interior Angles Converse

If same-side interior angles are supplementary, then the lines are parallel.



If $\angle 7 + \angle 8 = 180°$, then $r \parallel s$.

2 EXTERIOR ANGLE THEOREM

An **exterior angle** of a triangle is equal to the sum of the two remote interior angles.

Exterior angle = sum of two remote interior angles

$$\angle ext = \angle 1 + \angle 2$$



Example:

An exterior angle of a triangle measures 125° and one remote interior angle is 52°. Find the other remote interior angle.

Step 1: Set up the equation.

$125° = 52° + x$

Step 2: Solve.

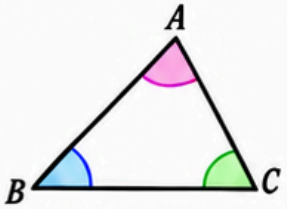
$x = 125° - 52° = 73°$

Answer: 73°

3 TRIANGLE ANGLE SUM THEOREM

The interior angles of a triangle always sum to 180°.

$$\angle A + \angle B + \angle C = 180°$$



Example:

In the triangle, $47° + 63° + x = 180°$.

Step 1: Add known angles.

$110° + x = 180°$

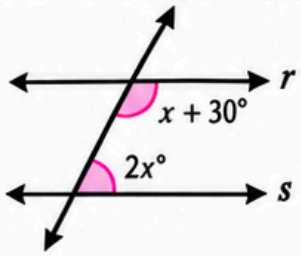
Step 2: Solve.

$x = 180° - 110° = 70°$

Answer: 70°

4 SAME-SIDE INTERIOR EXAMPLE

If same-side interior angles are supplementary, then the lines are parallel. Find the value of x and the angle measures.



Example:

$(x + 30)°$ and $(2x)°$ are supplementary.

Step 1: Set up the equation.

$(x + 30) + 2x = 180$

Step 2: Solve.

$3x = 150 \rightarrow x = 50$

Step 3: Find the angle measures.

$x + 30 = 80°$ and $2x = 100°$

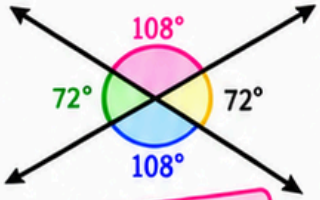
Answer: $x = 50$. Angles are 80° and 100°.

5 MULTI-STEP ANGLE CHASING

Use multiple angle relationships (corresponding, vertical, linear pair) to find unknown angles.

Example:

A corresponding angle is 108°. Find the measures of its vertical angle and its linear-pair adjacent angle.



Step 1: Corresponding angles are congruent, so the angle is 108°.

Step 2: Vertical angles are congruent, so the vertical angle is also 108°.

Step 3: A linear pair is supplementary, so the adjacent angle is $180° - 108° = 72°$.

Answer:
Vertical angle = 108°
Adjacent angle = 72°.

6 REASONS YOU MAY USE IN PROOFS

These are common reasons when working with angles and parallel lines.

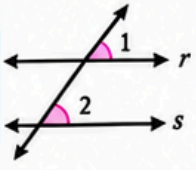
Vertical Angles Theorem	Vertical angles are congruent.
Corresponding Angles Postulate	If two parallel lines are cut by a transversal, then corresponding angles are congruent.
Alternate Interior Angles Theorem	If two parallel lines are cut by a transversal, then alternate interior angles are congruent.
Same-Side Interior Angles Theorem	If two parallel lines are cut by a transversal, then same-side interior angles are supplementary (their sum is 180°).
Linear Pair Postulate	If two angles form a linear pair, then they are supplementary (their sum is 180°).
Definition of Supplementary Angles	Two angles are supplementary if their measures add to 180°.

7 MINI PROOF IDEA

Use a converse to prove lines are parallel.

Given:

$\angle 1$ and $\angle 2$ are corresponding angles and are congruent ($\angle 1 \cong \angle 2$).



Prove:

$r \parallel s$

Reasoning:

Since corresponding angles are congruent, by the Converse of the Corresponding Angles Postulate, the lines are parallel.

8 QUICK REMINDERS & STRATEGY TIPS

DO REMEMBER:

- Identify the type of angle relationship (corresponding, alternate interior, alternate exterior, or same-side interior).
- Use the correct theorem or converse.
- Check if you are proving lines are parallel or using lines already known to be parallel.
- Label angles clearly in your diagram.
- Show your reasoning in a logical order.
- Make sure to write angle measures with °.
- Double-check your work and units.

COMMON MISTAKES:

- Using the wrong theorem (e.g., confusing corresponding and alternate interior).
- Assuming lines are parallel without proof.
- Using a converse when the direct theorem is needed, or vice versa.
- Mixing up interior and exterior angles.
- Not checking whether you are proving lines are parallel or using them as given.
- Forgetting that same-side interior angles are supplementary, not congruent.
- Arithmetic errors when solving for angles.